

Numerical:

2019 - Fall

1. b) Consider a non-interlaced raster monitor with a resolution of 1280×1024 . If horizontal and vertical retrace times are 20 microsecond each and then calculate the refresh fraction of the total refresh time per frame spend in retrace of the electron beam? Assume refresh rate of 60 frames per second.

so

Given :-

Resolution = 1280×1024 Horizontal retrace time, $t_h = 20$ microsecondVertical retrace time, $t_v = 20$ microsecondRefresh rate, $r = 60$ fps.

We know,

$$\text{Refresh rate} = 60 \text{ Hz} = \frac{1}{60} \text{ sec} = 0.0167 \text{ sec}$$

$$= 16.7 \text{ ms}$$

Now,

$$\text{Time for horizontal retrace} = 1024 \times 20 \text{ } \mu\text{s}$$

$$\text{Time for vertical retrace} = 20 \text{ } \mu\text{s}$$

$$\begin{aligned} \text{Total time spent for retrace} &= 1024 \times 20 + 20 \\ &= 20500 \text{ } \mu\text{s} = 20.5 \text{ ms} \end{aligned}$$

$$\therefore \text{The fraction of the total refresh time frame spent in retrace} = \frac{20.5}{16.7} = 1.227$$

Alternative,

$$\begin{aligned} \text{Total time spent for retrace} &= (m \times t_h) + t_v \\ &= (1024 \times 20) + 20 \\ &= 20500 \text{ } \mu\text{s} = 20.5 \text{ ms} \end{aligned}$$

2018-Spring:

1.6) Calculate the time access time for a pixel and a row of for a graphics algorithm system having resolution of 1024×640 and frequency of 60 Hz.

sol Given:

Resolution = 1024×640 pixels,

frequency, $f = 60$ Hz

$\therefore$ Refresh rate = ~~60 pixel/sec~~ 60 Hz = 60 fps

$\therefore$ Access time = $1024 \times 640 \times 60$

In 1 sec, 60 frames can be accessed.

Total pixel accessed per second

$$= 1024 \times 640 \times 60$$

$$= 39321600$$

$\therefore$ Access time per pixel = $\frac{1}{39321600}$

$$= 25.43 \text{ ns}$$

$$= 25.43 \text{ ns}$$

Refresh
rate

2016 - Spring:

1. b) A laser printer is capable of printing two pages (size 9×11 inch) per second at resolution of 600 pixels per inch. How many bits per second does such device required? (Assume 1 pixel = n bits) ?

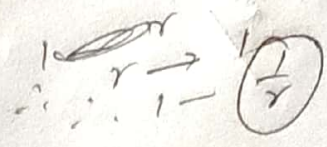
~~Q. 1~~ Given:

Storage required per page in pixel = $9 \times 11 \text{ inch}^2$
 $= 9 \times 11 \times 600 \text{ pixel} \text{ sec}$
 $(\because 1 \text{ inch} = 60 \text{ pixel})$

~~1 inch~~

Since the printer prints 2 pages per sec

$$\begin{aligned}\therefore \text{Required bits per second} &= 2 \times 9 \times 11 \times 600 \text{ pixel sec} \\ &= 118800 \text{ pixel sec} \\ &= 118800 n \text{ bits/sec} \\ &= 1.188 \times 10^5 n \text{ bits/sec}\end{aligned}$$



2012 - Fall

2. b) Consider a non-interlaced raster monitor with a resolution of n by m (n scan lines and m pixels per scan line), a refresh rate of r frames per second, a horizontal retrace time of t_h and a vertical retrace time of t_v . What is the fraction of the total refresh time per frame spent in retrace of the electron beam?

~~Total refresh time = $(m \times t_h) + t_v$~~

~~Fraction of the total refresh time per frame =~~

~~Refresh time =~~

Refresh time = $r \text{ fps} = \frac{1}{r} \text{ sec}$

Total time spent for retrace = $m \times t_h + t_v$

$\therefore$ Fraction of the total refresh time per frame spent in retrace =

$$\frac{1}{\frac{1}{r} \times (m \times t_h + t_v)} = \frac{r}{m \times t_h + t_v}$$

2017-Spring

2. a) Suppose RGB raster system to be designed using on 8 inch x 10 inch screen with a resolution of 100 pixels per inch in each direction. If we want to store 6 bits per pixel in the frame buffer, how much storage (in bytes) do we need for frame buffer?

solⁿ Given:

Resolution of system = $8 \times 10 \times 100 \times 100$ pixels

Total memory for frame buffer if we store 6 bits
bits = $6 \times 8 \times 100 \times 100 \times 100$ bits

$$\text{Total memory (in bytes)} = \frac{6 \times 8 \times 100 \times 100 \times 100}{8}$$

$$= 60000 \text{ bytes}$$

2018-Fall

2. a) How long would it take to load a 640×480 frame buffer with 12 bits per pixel, if 10^5 bits can be transferred per second? How long would it take to load 24 bits per pixel frame buffer with a resolution of 1280×1024 with using the same transfer rate?

solⁿ

For 1st system:

Resolution = 640×480 pixels

Total memory for frame buffer = $640 \times 480 \times 12$ bits

Since 10^5 bits takes 1 sec to transfer.

$\therefore 640 \times 480 \times 12$ bits take 1 sec

$$= \frac{640 \times 480 \times 12}{10^5} = 36.86 \text{ sec}$$

For 2nd 2nd second system:

Resolution = 1280×1024 pixel

Total memory for frame buffer = $24 \times 1280 \times 1024$ bits

Since bits transfer is same,

$$\therefore 24 \times 1280 \times 1024 \text{ bits take} = \frac{24 \times 1024 \times 1280}{10^5} \text{ sec}$$
$$= 314.57 \text{ sec}$$

Numericals of Ellipse drawing algorithm:

Q Draw an ellipse with radius w.r.t x-axis is 8 and radius w.r.t y-axis is 6.

Sol, Here,

Radius w.r.t x-axis, $r_x = 8$

Radius w.r.t y-axis, $r_y = 6$

$$P_0 = r_y^2 + (1/4 - r_y) r_x^2$$

For region 1: We have to stop calculation when

$$2x_{k+1}r_y^2 \geq 2y_{k+1}r_x^2$$

x_k	y_k	P_k	x_{k+1}	y_{k+1}	$2x_{k+1}r_y^2$	$2y_{k+1}r_x^2$
0	6	$P_0 = 6^2 + (1/4 - 6) \cdot 8^2 = -332$	1	6	72	768
1	6	$P_k = -332 + 2(0+1)6^2 + 6^2 + (6^2 - 6^2)8^2 - (6-6)64 = -224$	2	6	144	768
2	6	$P_k = -224 + 2(1+1)6^2 + 6^2 + (6^2 - 6^2) \cdot 8^2 - (6-6)64 = -44$	3	5	216	768
3	6	$P_k = -44 + 2(2+1)6^2 + 6^2 + (6^2 - 6^2) \cdot 8^2 - (6-6)8^2 = 208$	4	5	288	640
4	5	$P_k = 208 + 2(3+1)6^2 + 6^2 + (5^2 - 6^2) \cdot 8^2 - (5-6)8^2 = -108$	5	5	360	640
5	5	$P_k = -108 + 2(4+1)6^2 + 6^2 + (5^2 - 5^2)8^2 - (5-5)8^2 = 288$	6	4	432	512
6	4	$P_k = 288 + 2(5+1)6^2 + 6^2 + (4^2 - 5^2) \cdot 8^2 - (4-5)8^2 = 244$	7	3	504	384

For octant,

we have to stop calculation since

$$2x_{k+1}r_y^2 = 504 > 2y_{k+1}r_x^2 (=384)$$

For region 2:

x_k	y_k	P_{2k}	x_{k+1}	y_{k+1}
7	3	$P_{20} = (7+8)^2 \cdot 6^2 + (3+1)^2 \cdot 8^2 - 8^2 \cdot 6^2 = -23$	8	2
8	2	$P_{2k} = -23 + 8^2 - 2(3-1)8^2 + (8^2 - 7^2) \cdot 6^2$ $+ (8-7)6^2 = 361$	8	1
9	1	$P_{2k} = 361 + 8^2 - 2(2-1)8^2 + (8^2 - 8^2) \cdot 6^2 + (8-8)6^2$ $= 257$	8	0

Now,

$O_1(x, y)$	$O_2(x, -y)$	$O_3(-x, -y)$	$O_4(-x, y)$
(0, 6)	(0, -6)	(0, -6)	(0, 6)
(1, 6)	(1, -6)	(-1, -6)	(-1, 6)
(2, 6)	(2, -6)	(-2, -6)	(-2, 6)
(3, 6)	(3, -6)	(-3, -6)	(-3, 6)
(4, 5)	(4, -5)	(-4, -5)	(-4, 6)
(5, 5)	(5, -5)	(-5, -5)	(-5, 5)
(6, 4)	(6, -4)	(-6, -4)	(-6, 4)
(7, 3)	(7, -3)	(-7, -3)	(-7, 3)
(8, 2)	(8, -2)	(-8, -2)	(-8, 2)
(8, 1)	(8, -1)	(-8, -1)	(-8, 1)
(8, 0)	(8, 0)	(-8, 0)	(-8, 0)

Boundary Fill Algorithm:

- In boundary fill algorithm, area filling is started inside a point within a boundary region and fill the region with the specified color until it reaches the boundary.
- Boundary fill algorithm is used in interactive packages where we can specify the region boundary.
- Boundary fill algorithm is less time consuming.
- Boundary fill algorithm searches for boundary.

Flood Fill Algorithm:

- In ~~Area~~ Flood fill algorithm, area filling is started from a point and it replaces the old color with the new color.
- Flood fill algorithm is used ~~where~~ where we cannot specify the boundary.
- Flood fill algorithm is ~~more~~ ^{more} time consuming.
- Flood fill algorithm searches for old color.

2013- Spring

1) Scale a triangle $A(0,0)$, $B(1,1)$ & $C(3,2)$ by twice its original size about origin and fixed point $(-1,-2)$.

solⁿ Given points:

$A(0,0)$, $B(1,1)$ & $C(3,2)$

$$\therefore S_x = S_y = 2$$

i) Scale about origin

$$P' = S(S_x, S_y) \cdot P$$

$$= S(2, 2) \cdot P$$

$$P = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 3 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\therefore P' = \begin{bmatrix} 0 & 2 & 6 \\ 0 & 2 & 4 \\ 1 & 1 & 1 \end{bmatrix}$$

Thus the scaled point about the origin are:

$A(0,0)$, $B(2,2)$ & $C(6,4)$.

ii) Scale about fixed point $(-1,-2)$

$$P' = T(-1,-2) \cdot S(2,2) \cdot T(1,2) \cdot P$$

$$= \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 3 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 0 & -1 \\ 0 & 2 & -2 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 4 \\ 2 & 3 & 4 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\therefore P' = \begin{bmatrix} 1 & 3 & 7 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$$

$\therefore$ The scaled point about the fixed fixed points $(4, -2)$ are: $A(1, 2)$, $B(3, 4)$ & $C(7, 6)$

Raster Scan display

- has low or poor resolution
- cheaper than random scan system.
- Pixels are used to draw an image.
- Uses interlacing
- Refresh area is independent of picture complexity.
- Produce jagged line
- Editing is difficult.
- Examples: CRT, TV, Printer etc

Random Scan display

- higher resolution
- expensive than raster scan
- Mathematic function are used to draw an image
- It's not use interlacing
- Refresh area depends on picture complexity.
- produce smooth line
- Editing is easy.
- Examples: Pen plotter etc

1. Suppose RGB raster system to be designed using on 8 inch x 10 inch screen with a resolution of 100 pixels per inch in each direction. If we want to store 6 bits per pixel in the frame buffer, how much storage (in bytes) do we need from the frame buffer? (2017-sp)

Ans Here,

$$\text{Resolution of system} = (8 \times 10) \text{ inch} \times 100 \text{ pixels per inch} \\ = 8 \times 10^5 \text{ pixels}$$

$$\begin{aligned} \text{Total memory for frame buffer if we store 6 bits} \\ &= 8 \times 100 \times 10 \times 100 \times 6 \text{ bits} \\ &= 8 \times 10^5 \times 6 \text{ bits} \\ &= 56 \times 10^5 \text{ bits} \end{aligned}$$

$$\begin{aligned} \therefore \text{Memory used in bytes} &= \frac{56 \times 10^5}{8} \text{ bytes } (\because 8 \text{ bit} = 1 \text{ byte}) \\ &= 6 \times 10^5 \text{ bytes.} \end{aligned}$$

2. How long would it take to load a 640×480 frame buffer with 12 bits per pixel if 10^5 bits can be transferred per second? How long would it take to load 24 bits per pixel frame buffer with a resolution of 1280×1024 using the same transfer rate?

Given:-

For 1st system:

$$\text{Resolution} = 640 \times 480 \text{ pixels}$$

$$\begin{aligned} \text{Total memory for frame buffer if we store 12 bits} \\ \text{per pixel} &= 640 \times 480 \times 12 \text{ bits} \end{aligned}$$

$\therefore 10^5$ bits can be transferred in 1 sec.

$\Rightarrow 1$ bit can be transferred in $\frac{1}{10^5}$ sec.

$\Rightarrow 640 \times 480 \times 12$ bits can be transferred in $\frac{640 \times 480 \times 12}{10^5}$ sec.
 $= 36.864$ sec.

For 2nd System:

Resolution = 1280×1024 pixels.

Total memory for frame buffer with 24 bits per pixel = $1280 \times 1024 \times 24$ bits.

Since the transfer rate is same, so,

$1280 \times 1024 \times 24$ bits can be transferred in
 $\frac{1280 \times 1024 \times 24}{10^5} = 314.5728$ sec.

3. Consider two raster system with a resolution of 640×480 and 1280×1024 . How many pixels could be accessed per second in each of these system by a display controller that refreshes the screen at a rate of 60 frames per second? What is the access time per pixel in each system?

sol

For ~~1st~~ 640×480 system:

Resolution = 640×480 pixels

Refresh rate = 60 frames per sec.

In 1 sec, 60 frames can be accessed,
Since 1 frame means total no. of pixels.

$$\begin{aligned} \therefore \text{In 1 sec, } 60 \times \text{Resolution} & \text{ pixels accessed per second} \\ &= 60 \times \text{Resolution} \\ &= 60 \times 480 \times 640 \\ &= 1843200 \text{ pixels.} \end{aligned}$$

$$\text{Access time per pixel} = \frac{1}{1843200} = 5.42 \times 10^{-8} \text{ sec}$$

For second system:

$$\text{Resolution} = 1280 \times 1024$$

$$\text{Refresh rate} = 60 \text{ frames per sec}$$

$$\begin{aligned} \therefore \text{Total pixels accessed per second} &= 1280 \times 1024 \times 60 \\ &= 78643200 \text{ pixels.} \end{aligned}$$

$$\text{Access time per pixel} = \frac{1}{78643200} = 1.27 \times 10^{-8} \text{ sec.}$$

Numericals 2012 - Spring

Q. Calculate the view transformation matrix with the given transformation.

- Given triangle with vertices: $A(5, 5)$, $B(15, 5)$, $C(10, 10)$
- Given window co-ordinate: $(7, 4)$, $(13, 4)$, $(13, 8)$, $(7, 8)$
- Given viewport coordinate: $(17, 4)$, $(18, 7)$, $(18, 8)$, $(17, 8)$

Sol

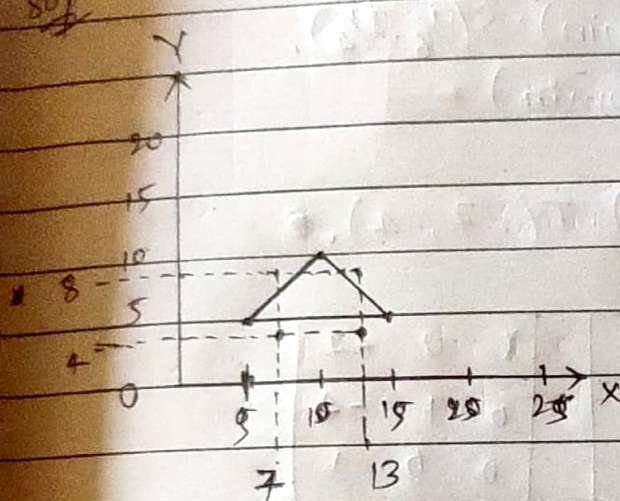


Fig. window coordinate

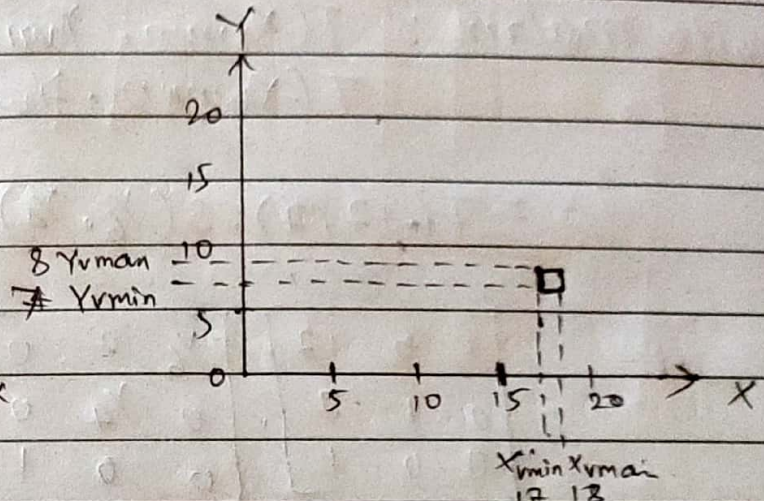


Fig. viewport coordinate

Here,

$$X_{wmin} = 7, X_{wmax} = 13$$

$$Y_{wmin} = 4, Y_{wmax} = 8$$

Here,

$$X_{vmin} = 17, X_{vmax} = 18$$

$$Y_{vmin} = 7, Y_{vmax} = 8$$

Now,

$$S_x = \frac{X_{vmax} - X_{vmin}}{X_{wmax} - X_{wmin}} = \frac{18 - 17}{13 - 7} = \frac{1}{6}$$

$$S_y = \frac{Y_{vmax} - Y_{vmin}}{Y_{wmax} - Y_{wmin}} = \frac{8 - 7}{8 - 4} = \frac{1}{4}$$

~~COM = T~~

- i) Transform wc to its origin. i.e. $T(x_{wmin}, -y_{wmin})$
- ii) Scale it by $s_x = 1/6$ & $s_y = 1/4$ i.e. $S(1/6, 1/4)$
- iii) Transform it to the relative position into the viewport i.e. $T(x_{vmin}, y_{vmin})$.

$$\text{Composite matrix} = T(x_{vmin}, y_{vmin}) \cdot S(1/6, 1/4) \cdot T(x_{wmin}, -y_{wmin})$$

$$= T(-17, 7) \cdot S(1/6, 1/4) \cdot T(-7, -4) \cdot P$$

$$= \begin{bmatrix} 1 & 0 & 17 \\ 0 & 1 & 7 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1/6 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -7 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore \text{COM} = \begin{bmatrix} 0.17 & 0 & 15.83 \\ 0 & 0.25 & 6 \\ 0 & 0 & 1 \end{bmatrix}$$

Now

$$P' = \text{COM} \cdot P$$

$$= \begin{bmatrix} 0.17 & 0 & 15.83 \\ 0 & 0.25 & 6 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 5 & 15 & 10 \\ 5 & 5 & 10 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\therefore P' = \begin{bmatrix} 16.67 & 18.33 & 17.5 \\ 7.25 & 7.25 & 8.5 \\ 1 & 1 & 1 \end{bmatrix} \quad \underline{\text{Ans}}$$

- ②. Find the normalization that maps a window whose lower left corner is at (1,1) and upper left right corner is at (3,5) on to viewport that is entire normalized device screen.

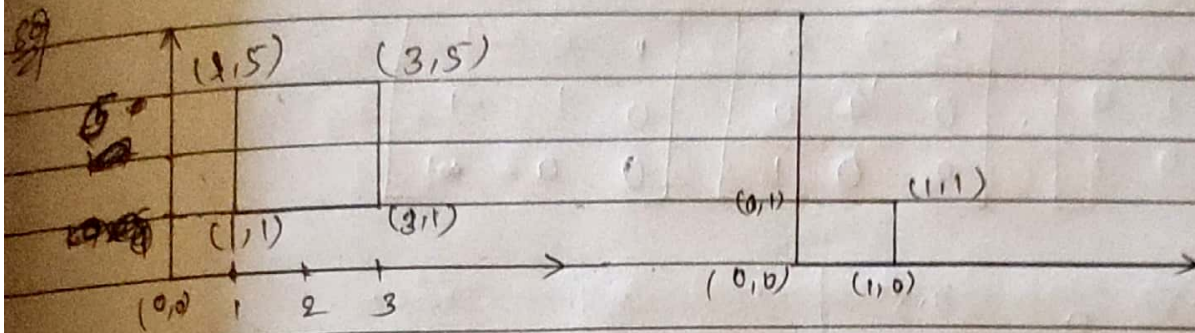


fig. window coordinate.

fig. viewport coordinate

Here,

$$X_{wmin} = 1; X_{wmax} = 3$$

$$X_{vmin} = 0; X_{vmax} = 1$$

$$Y_{wmin} = 1; Y_{wmax} = 5$$

$$Y_{vmin} = 0; Y_{vmax} = 1$$

Now,

$$S_x = \frac{X_{vmax} - X_{vmin}}{X_{wmax} - X_{wmin}} = \frac{1 - 0}{3 - 1} = \frac{1}{2}$$

$$S_y = \frac{Y_{vmax} - Y_{vmin}}{Y_{wmax} - Y_{wmin}} = \frac{1 - 0}{5 - 1} = \frac{1}{4}$$

The steps to calculate view port transformation.

i) Transform window coordinate to its origine. Φ

i.e. $T(-X_{wmin}, -Y_{wmin})$

ii) Scale it by S_x & S_y i.e. $S(S_x, S_y)$

iii) Transform it to the same relative position to in viewport

i.e. $T(X_{vmin}, Y_{vmin})$

Now,

$$COM = T(x_{vmin}, y_{vmin}) \cdot S(s_x, s_y) \cdot T(-x_{wmin}, -y_{wmin})$$

$$= T(0, 0) \cdot S(1/2, 1/4) \cdot T(-1, -1)$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5 & 0 & -0.5 \\ 0 & 0.25 & -0.25 \\ 0 & 0 & 1 \end{bmatrix} \quad \underline{Ans}$$

2016-Fall

Q.6) Consider two raster systems with resolution of 640 by 480 and 1280 by 1024. How many pixels could be accessed per second in each of these systems by a display controller that refreshes the screen at a rate of 60 frames per second? What is the access time per pixel in each system?

Given:

for 1st system:

Resolution = 640 x 480 pixels

Refresh rate = 60 fps

Total no. of pixels accessed per second
= 60 x 640 x 480 pixels

$$\therefore \text{Total Access time per pixel} = \frac{1}{60 \times 640 \times 480} = 5.425 \times 10^{-8} \text{ sec}$$

for 2nd system:

Resolution = 1280 x 1024 pixels

Refresh rate = 60 fps

Total no. of pixels accessed per second = 60 x 1280 x 1024

$$\therefore \text{Total access time per pixel} = \frac{1}{60 \times 1280 \times 1024} = 1.271 \times 10^{-8} \text{ sec}$$

2015-Spring

3.6) What will be the final coordinates of a triangle with vertices $A(2,3)$, $B(3,3)$, $C(3,2)$ after rotating it α by 45° in anticlockwise direction then shifting it down by 3 units and finally enlarging it by twice its original size?

Sol
The given vertices of a triangle are:
 $A(2,3)$, $B(3,3)$ & $C(3,2)$.

i) Rotate by 45° i.e. $R^A(45)$

ii) shifting it down by 3 units i.e. $t_y = 3$

iii) enlarging it by twice its original size
i.e. $S_x = S_y = 2$

Now

$$COM = S(2,2) \cdot T(0,-3) \cdot R^A(45^\circ)$$

$$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos 45 & -\sin 45 & 0 \\ \sin 45 & \cos 45 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore COM = \begin{bmatrix} 1.4 & -1.4 & 0 \\ 1.4 & 1.4 & -6 \\ 0 & 0 & 1 \end{bmatrix}$$

Note

$$P' = COM \times P$$

$$= \begin{bmatrix} 1.4 & -1.4 & 0 \\ 1.4 & 1.4 & -6 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 3 \\ 3 & 3 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\therefore P' = \begin{bmatrix} -1.4 & 0 & -1.4 \\ 1.1 & 2.5 & 1.1 \\ 1 & 1 & 1 \end{bmatrix}$$

Thus the final co-ordinates are:

A (-1.4, 1.1), B (0, 2.5) & C (-1.4, 1.1) Ans

Gouraud shading v/s Phong shading

S.N.	Gouraud shading	Phong shading
i>	Gouraud shading was introduced by Henri Gouraud in 1970.	Phong shading was introduced by Phong Bui Tuong in 1973.
ii>	Gouraud shading evaluates the illumination at border vertices and interpolates.	Phong shading evaluates the illumination at every point of the polygon surfaces.
iii>	Gouraud shading interpolates colors along edges and scanline.	Phong shading interpolates normals instead of colors.
iv>	Gouraud shading is cheaper than phong shading.	Phong shading is expensive than Gouraud shading.
v>	In Gouraud shading, lighting equation is used at each vertex.	In Phong shading lighting equation is used at each pixel.
vi>	Gouraud shading gives less accurate results.	Phong shading gives more accurate results.
vii>	Gouraud shading produces average quality.	Phong shading produces good quality.
viii>	Gouraud shading requires average processing and less time.	Phong shading requires high processing and more time.

Chapter-7

Illumination & Shading

→ Ambient light is a component of lighting in computer graphics used to simulate the overall indirect illumination present in a scene.

SN.	Diffuse Reflection	Specular Reflection
i>	Diffuse reflection is produced from rough or matte surfaces.	Specular reflection is produced from smooth or glossy surface.
ii>	In diffuse reflection, light scatters in all direction.	In specular reflection, light reflects in a concentrated direction.
iii>	In diffuse reflection, incident angle = reflected angle.	In specular reflection, incident angle = reflection angle.
iv>	Diffuse reflection is visible from a wide range of the light source angles.	Specular reflection is visible only from specific angles.
v>	It doesnot create a clear reflection of light source.	It creates a clear reflection of light source.
vi>	Examples: paper, fabric, non-polished wood reflection etc.	Examples: mirror, polished metal, glass reflections etc.

Z-buffer v/s A-buffer method:

S.N.	Z-buffer	A-buffer
i>	Z-buffer is also called depth buffer method.	A-buffer method is also called accumulation buffer method.
ii>	Z-buffer method stores per pixel depth values.	A-buffer stores method stores per pixel fragment details.
iii>	A Z-buffer method requires more memory for depth buffer only.	A-buffer method requires memory for multiple buffers.
iv>	Z-buffer handles occlusion naturally.	A-buffer requires additional processing to handle occlusion.
v>	Z-buffer has limited support for transparent objects.	A-buffer has unlimited support for transparent objects.

chapter-6

visible surface Detection.

→ The surfaces which are hidden and blocked from view are called hidden surfaces.

→ The identification and removal of hidden surfaces is called hidden surface removal.

object Space method.	Image Space Method
i> OSM operates directly on object geometry.	ISM operates on the rendered image or screen.
ii> OSM performs visibility tests on objects directly.	ISM performs visibility test on comparing pixel values.
iii> OSM is efficient for static or non-deforming objects.	ISM is efficient for dynamic or deforming objects.
iv> OSM has simpler algorithms & calculations.	ISM has complex algorithms and calculations.
v> OSM handles limited occlusion within objects.	ISM handles inter-object and pixel-level occlusion.
vi> OSM are not well suited for handling transparent objects.	ISM are well suited for handling transparent objects.
vii> For example: Back-face detection method.	For example: Depth buffer method, Scan-line method etc.

For $|m| < 1$; Δx times steps
 For $|m| > 1$; Δy times steps

Numericals:

1. Digitize a line with end points A(2,2) and B(12,10) using BLA.

sol The givens of a line are:
 A(2,2) & B(12,10)

$$\text{slope, } m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{10 - 2}{12 - 2} = \frac{8}{10} = \frac{4}{5} < 1$$

$$\therefore \Delta x = 10; \Delta y = 8$$

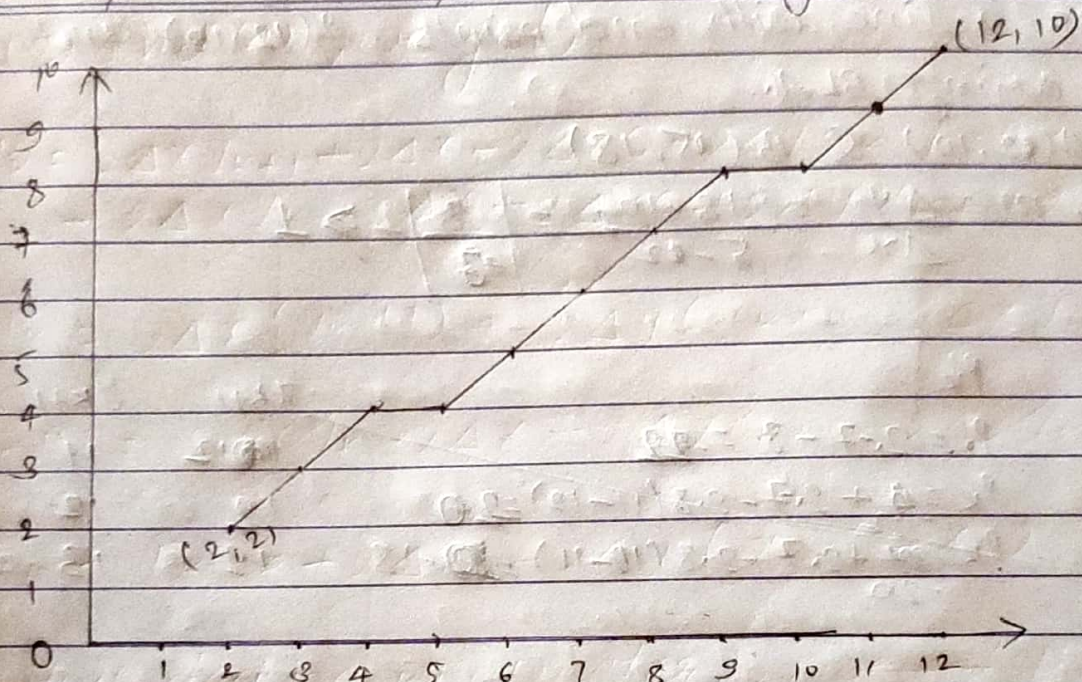
we stop the calculation when we reach at end point.

S.N.	x_k	y_k	P_k	x_{k+1}	y_{k+1}
1.	2	2	$P_0 = 2 \cdot 10 - 10 = 10 > 0$	3	3
2.	3	3	$P_k = 6 + 2 \cdot 8 - 2 \cdot 10(3 - 2) = 2 > 0$	4	4
3.	4	4	$P_k = 2 + 2 \cdot 8 - 2 \cdot 10(4 - 3) = -2 < 0$	5	4
4.	5	4	$P_k = -2 + 2 \cdot 8 - 2 \cdot 10(4 - 4) = 14 > 0$	6	5
5.	6	5	$P_k = 14 + 2 \cdot 8 - 2 \cdot 10(5 - 4) = 10 > 0$	7	6
6.	7	6	$P_k = 10 + 2 \cdot 8 - 2 \cdot 10(6 - 5) = 6 > 0$	8	7
7.	8	7	$P_k = 6 + 2 \cdot 8 - 2 \cdot 10(7 - 6) = 2 > 0$	9	8
8.	9	8	$P_k = 2 + 2 \cdot 8 - 2 \cdot 10(8 - 7) = -2 < 0$	10	8
9.	10	8	$P_k = -2 + 2 \cdot 8 - 2 \cdot 10(8 - 8) = 14 > 0$	11	9
10.	11	9	$P_k = 14 + 2 \cdot 8 - 2 \cdot 10(9 - 8) = 10 > 0$	12	10

we have to stop calculation.

The intermediate points are: (2,2), (3,3), (4,4), (5,4), (6,5), (7,6), (8,7), (9,8), (10,8), (11,9) & (12,10).

we have plotted these points in the graph paper,



2) Digitize a line with end points $A(2, 10)$ and $B(5, 18)$ using BLA.

Here, the given end points of the line are;
 $A(2, 10)$ & $B(5, 18)$

$$\therefore \text{Slope, } m = \Delta y / \Delta x = y_2 - y_1 / x_2 - x_1 = 18 - 10 / 5 - 2 = (8/3) >$$

S.N.	x_k	y_k	P_k	x_{k+1}	y_{k+1}
1.	12	10	$P_0 = 2.7 - 2.8 = -0.1 < 0$	11	11
2.	12	11	$P_k = 2.7 - 2.8(11 - 12) = 4.4 > 0$	10	12
3.	10	12	$P_k = 2.7 - 2.8(10 - 11) = 3.0 > 0$	9	13
4.	9	13	$P_k = 2.7 - 2.8(9 - 10) =$		

5.
6.
7.
8.

②. Digitize a line with end points A(12,10) and B(5,18) using BLA.

Sol Here, A(12,10) & B(5,18)

$$\therefore \text{slope, } m = \frac{\Delta y}{\Delta x} = \frac{18-10}{5-12} = \frac{8}{-7} \Rightarrow \perp$$

S.N.	x_k	y_k	P_k	x_{k+1}	y_{k+1}
1.	12	10	$P_0 = 2 \cdot 7 - 8 = -2$	11	11
2.	11	11	$P_k = -2 + 2 \cdot 7 - 2 \cdot 8(11-12) = 2$	12	12
3.	10	12	$P_k = 2 + 2 \cdot 7 - 2 \cdot 8(10-11) = -18$	12	13

4.

5.

6.

7.

8.

S.N.	x_k	y_k	P_k	x_{k+1}	y_{k+1}
1.	2	10	$P_0 = 2 \cdot 3 - 8 = -2$	2	11
2.	2	11	$P_k = -2 + 2 \cdot 3 - 2 \cdot 8(2-2) = 4$	3	12
3.	3	12	$P_k = 4 + 2 \cdot 3 - 2 \cdot 8(3-2) = -6$	3	13
4.	3	13	$P_k = -6 + 2 \cdot 3 - 2 \cdot 8(3-3) = 0$	4	14
5.	4	14	$P_k = 0 + 2 \cdot 3 - 2 \cdot 8(4-3) = -10$	4	15
6.	4	15	$P_k = -10 + 2 \cdot 3 - 2 \cdot 8(4-4) = -4$	4	16
7.	4	16	$P_k = -4 + 2 \cdot 3 - 2 \cdot 8(4-4) = 2$	5	17
8.	5	17	$P_k = 2 + 2 \cdot 3 - 2 \cdot 8(5-4) = -8$	5	18

(B) Digitize the points $A(-2, -4)$ & $B(-6, -9)$ using BLA.

sol Here $A(-2, -4)$ & $B(-6, -9)$

$$\therefore \text{slope, } m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-9 - (-4)}{-6 - (-2)} = \frac{-5}{-4}$$

$$\therefore |m| = \left| \frac{5}{4} \right| > 1$$

$$\therefore P_0 = 2\Delta x - \Delta y$$

S.N.	x_k	y_k	P_k	x_{k+1}	y_{k+1}
1.	-2	-4	$P_0 = 2 \cdot 4 - 5 = 3$	-3	-5
2.	-3	-5	$P_k = 3 + 2 \cdot 4 - 2 \cdot 5(3 - 2) = 1$	-4	-6
3.	-4	-6	$P_k = 1 + 2 \cdot 4 - 2 \cdot 5(4 - 3) = -1$	-4	-7
4.	-4	-7	$P_k = -1 + 2 \cdot 4 - 2 \cdot 5(4 - 4) = 7$	-5	-8
5.	-5	-8	$P_k = 7 + 2 \cdot 4 - 2 \cdot 5(5 - 4) = 5$	-5	-9

(A) Rasterize the points $A(2, 3)$ and $B(8, -8)$ using BLSOP.
Here, $A(2, 3)$ and $B(8, -8)$.

$$\therefore \text{slope, } m = \frac{-8-3}{8-2} = \left| \frac{-11}{6} \right| > 1$$

$$\therefore \Delta x = 6$$

$$\therefore \Delta y = 11$$

S.N.	x_k	y_k	P_k	x_{k+1}	y_{k+1}
1.	2	3	$P_0 = 2 \cdot 6 - 11 = 1$	3	2 2
2.	3	2	$P_k = 1 + 2 \cdot 6 - 2 \cdot 11 (3-2) = -9$	3	1
3.	3	1	$P_k = -9 + 2 \cdot 6 - 2 \cdot 11 (3-3) = 3$	4	0
4.	4	0	$P_k = 3 + 2 \cdot 6 - 2 \cdot 11 (4-3) = -7$	4	-1
5.	4	-1	$P_k = -7 + 2 \cdot 6 - 2 \cdot 11 (4-4) = 5$	5	-2
6.	5	-2	$P_k = 5 + 2 \cdot 6 - 2 \cdot 11 (5-4) = -5$	5	-3
7.	5	-3	$P_k = -5 + 2 \cdot 6 - 2 \cdot 11 (5-5) = 7$	6	-4
8.	6	-4	$P_k = 7 + 2 \cdot 6 - 2 \cdot 11 (6-5) = -3$	6	-5
9.	6	-5	$P_k = -3 + 2 \cdot 6 - 2 \cdot 11 (6-6) = 9$	7	-6
10.	7	-6	$P_k = 9 + 2 \cdot 6 - 2 \cdot 11 (7-6) = -1$	7	-7
11.	7	-7	$P_k = -1 + 2 \cdot 6 - 2 \cdot 11 (7-7) = 11$	8	-8

* Numericals:

1. Draw a Circle with radius 8 using mid-point circle algorithm.

sol: Here, radius, $r = 8$, centre $= (0, 0)$

$\therefore$ decision algorithm, $P_0 = 1 - r$

$\therefore$ The start point will be $(0, r)$.

We have to stop calculating intermediate points at

$$x_{k+1} \geq y_{k+1}$$

x_k	y_k	P_k	x_{k+1}	y_{k+1}	Plot-Point
0	8	$P_0 = 1 - 8 = -7$	1	8	(1, 8)
1	8	$P_k = -7 + 2(0+1) + (8^2 - 8^2) - (8-8) + 1 = -4$	2	8	(2, 8)
2	8	$P_k = -4 + 2(1+1) + (8^2 - 8^2) - (8-8) + 1 = 1$	3	7	(3, 7)
3	7	$P_k = 1 + 2(2+1) + (7^2 - 8^2) - (7-8) + 1 = -6$	4	7	(4, 7)
4	7	$P_k = -6 + 2(3+1) + (7^2 - 7^2) - (7-7) + 1 = 3$	5	6	(5, 6)
5	6	$P_k = 3 + 2(4+1) + (6^2 - 7^2) - (6-7) + 1 = 2$	6	5	(6, 5)

We stopped calculation i.e. $6 \geq 5$. So, the intermediate points are:

	$Q_1(x, y)$	$Q_2(x, -y)$	$Q_3(-x, -y)$	$Q_4(-x, y)$
	(0, 8)	(0, -8)	(0, -8)	(0, 8)
5	(1, 8)	(1, -8)	(-1, -8)	(-1, 8)
	(2, 8)	(2, -8)	(-2, -8)	(-2, 8)
	(3, 7)	(3, -7)	(-3, -7)	(-3, 7)
	(4, 7)	(4, -7)	(-4, -7)	(-4, 7)
5	(5, 6)	(5, -6)	(-5, -6)	(-5, 6)
1	(6, 5)	(6, -5)	(-6, -5)	(-6, 5)
2				
3	(5, 6)	(5, -6)	(-5, -6)	(-5, 6)
	(6, 5)	(6, -5)	(-6, -5)	(-6, 5)
5	(7, 4)	(7, -4)	(-7, -4)	(-7, 4)
	(7, 3)	(7, -3)	(-7, -3)	(-7, 3)
	(8, 2)	(8, -2)	(-8, -2)	(-8, 2)
	(8, 1)	(8, -1)	(-8, -1)	(-8, 1)
	(8, 0)	(8, 0)	(-8, 0)	(-8, 0)

Q. Digitize a circle with radius 10 and centre (10, 20) using mid-point circle algorithm.

Here,

radius, $r = 10$

centre = (10, 20)

$$\therefore P_0 = 1 - r = 1 - 10 = -9$$

The start point will be $(0, r) = (0, 10)$.

x_k	y_k	P_k	x_{k+1}	y_{k+1}	Plot-point
0	10	$P_0 = 1 - 10 = -9$	1	10	(11, 30)
1	10	$P_k = -9 + 2(0+1) + (10^2 - 10^2) - (10 - 10) + 1$ $= -6$	2	10	(12, 30)
2	10	$P_k = -6 + 2(1+1) + (10^2 - 10^2) - (10 - 10) + 1$ $= -1$	3	10	(13, 30)
3	10	$P_k = -1 + 2(2+1) + (10^2 - 10^2) - (10 - 10) + 1$ $= 6$	4	9	(14, 29)
4	9	$P_k = 6 + 2(3+1) + (9^2 - 10^2) - (9 - 10) + 1$ $= -3$	5	9	(15, 29)
5	9	$P_k = -3 + 2(4+1) + (9^2 - 10^2) - (9 - 9) + 1$ $= 8$	6	8	(16, 28)
6	8	$P_k = 8 + 2(5+1) + (8^2 - 9^2) - (8 - 9) + 1$ $= 5$	7	7	(17, 27)

We stop calculation as $x_{k+1} \geq x_{k+1}$. So the intermediate points are:

x_k	y_k	p_k	x_{k+1}
$Q_1(x, y)$	$Q_2(x, y)$	$Q_3(-x, -y)$	$Q_4(-x, y)$
$(10, 30)$	$(10, -30)$	$(-10, -30)$	$(-10, 30)$
$(11, 30)$	$(11, -30)$	$(-11, -30)$	$(-11, 30)$
$(12, 30)$	$(12, -30)$	$(-12, -30)$	$(-12, 30)$
$(13, 30)$	$(13, -30)$	$(-13, -30)$	$(-13, 30)$
$(14, 29)$	$(14, -29)$	$(-14, -29)$	$(-14, 29)$
$(15, 29)$	$(15, -29)$	$(-15, -29)$	$(-15, 29)$
$(16, 28)$	$(16, -28)$	$(-16, -28)$	$(-16, 28)$
$(17, 27)$	$(17, -27)$	$(-17, -27)$	$(-17, 27)$
$(27, 17)$	$(27, -17)$	$(-27, -17)$	$(-27, 17)$
$(28, 16)$	$(28, -16)$	$(-28, -16)$	$(-28, 16)$
$(29, 15)$	$(29, -15)$	$(-29, -15)$	$(-29, 15)$
$(29, 14)$	$(29, -14)$	$(-29, -14)$	$(-29, 14)$
$(30, 13)$	$(30, -13)$	$(-30, -13)$	$(-30, 13)$
$(30, 12)$	$(30, -12)$	$(-30, -12)$	$(-30, 12)$
$(30, 11)$	$(30, -11)$	$(-30, -11)$	$(-30, 11)$
$(30, 10)$	$(30, -10)$	$(-30, -10)$	$(-30, 10)$

* Numericals:

Q. Draw an ellipse with radius w.r.t. x-axis (r_x) = 8 and w.r.t. y-axis (r_y) = 6.

sol. Here, $r_x = 8$ & $r_y = 6$.

$$\therefore P_0 = r_y^2 + \left(\frac{1}{4} - r_y\right) r_x^2$$

For first octant: We have to stop calculating points when $2x_{k+1}r_y^2 \geq 2y_{k+1}r_x^2$

x_k	y_k	P_k	x_{k+1}	y_{k+1}	$2x_{k+1}r_y^2$	$2y_{k+1}r_x^2$
0	6	$P_0 = 6^2 + (\frac{1}{4} - 6) \cdot 8^2 = -332$	1	6	72	768
1	6	$P_{1K} = -332 + 6^2 + 2(0+1)6^2 + (6^2 - 6^2)8^2$ $- (6-6)8^2 = -224$	2	6	144	768
2	6	$P_{1K} = -224 + 6^2 + 2(1+1)6^2 + (6^2 - 6^2)8^2$ $- (6-6)8^2 = -44$	3	6	216	768
3	6	$P_{1K} = -44 + 6^2 + 2(2+1)6^2 + (6^2 - 6^2)8^2$ $- (6-6)8^2 = 208$	4	5	288	640
4	5	$P_{1K} = 208 + 6^2 + 2(3+1)6^2 + (5^2 - 6^2)8^2$ $- (5-6)8^2 = -108$	5	5	360	640
5	5	$P_{1K} = -108 + 6^2 + 2(4+1)6^2 + (5^2 - 5^2)8^2$ $- (5-5)8^2 = 288$	6	4	432	512
6	4	$P_{1K} = 288 + 6^2 + 2(5+1)6^2 + (4^2 - 5^2)8^2$ $- (4-5)8^2 = 244$	7	3	504	384

We have to stop now i.e. $2x_{k+1}r_y^2 \geq 2y_{k+1}r_x^2$ for first octant.

For end octant:

x_k	y_k	p_{2k}	x_{k+1}	y_{k+1}
7	3	$p_{2k} = (7 + \frac{1}{2})^2 \times 6^2 + (3 + -1)^2 \times 8^2 - 6^2 \times 8^2 = -23$	8	2
8	2	$p_{2k} = (8 + \frac{1}{2})^2 \times 6^2 + (2 - 1)^2 \times 8^2 - 6^2 \times 8^2 = 361$	8	1
8	1	$p_{2k} = (8 + \frac{1}{2})^2 \times 6^2 + (2 - 1)^2 \times 8^2 - 6^2 \times 8^2 =$		

Numericals

2013-spring

- ①. Scale a triangle $A(0,0)$, $B(1,1)$ & $C(3,2)$ by twice its original size about origin and fixed point $(-1,-2)$.
sq. The given points of a triangle are:
 $A(0,0)$, $B(1,1)$ & $C(3,2)$

Since a triangle is scaled by twice its size,
i.e. $S_x = S_y = 2$

a) Scale about origin:

According to homogeneous coordinate.

$$P' = S(S_x, S_y) \cdot P$$

$$\Rightarrow P(x, y) = S(2, 2) \cdot P$$

$$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 3 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\therefore P(x, y) = \begin{bmatrix} 0 & 2 & 6 \\ 0 & 2 & 4 \\ 1 & 1 & 1 \end{bmatrix}$$

Thus the scaled points of a triangle about origin is
 $(A(0,0), B(2,2), C(6,4))$. Ans

Scale about point $(-1, -2)$.

Translate the fixed point to the origin i.e. $T(1, 2)$.

Scale with scaling parameter i.e. $S(2, 2)$.

Re-translate the fixed to the original place i.e. $T(-1, -2)$.

$$COM = T(-1, -2) \cdot S(2, 2) \cdot T(1, 2)$$

$$P' = COM \cdot P$$

$$= T(-1, -2) \cdot S(2, 2) \cdot T(1, 2) \cdot P$$

$$= \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 3 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & 4 & 8 \\ 4 & 6 & 8 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\therefore P' = \begin{bmatrix} 1 & 3 & 7 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$$

Thus the scaled points about the fixed point $(-1, -2)$ are:

$A(1, 2)$, $B(3, 4)$ & $C(7, 6)$. Ans

2014-fall

② Perform a 45° rotation of a line $A(5,3)$ and $B(10,15)$ about origin.

807 Here, $A(5,3)$ & $B(10,15)$

$$\theta = 45^\circ$$

~~According to~~ According to homogeneous coordinates:

$$P' = R(\theta) \cdot P$$

$$= R(45) \cdot P$$

$$= \begin{bmatrix} \cos(45) & -\sin(45) & 0 \\ \sin(45) & \cos(45) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 5 & 10 \\ 3 & 15 \\ 1 & 1 \end{bmatrix}$$

$$\therefore P' = \begin{bmatrix} 1.4 & -3.5 \\ 5.7 & 17.7 \\ 1 & 1 \end{bmatrix} \quad \#0$$

Thus the after rotation about 45° , the points are;
 $A(1.4, 5.7)$ and $B(-3.5, 17.7)$. A

Q. 14. Spring

perform a 45° rotation of a line $A(8,3)$ and $B(14,10)$

i) about origin

b) about fixed point $(4,2)$.

Here, $\theta = 45^\circ$

$A(8,3)$ & $B(14,10)$

a) About origin:

$$P' = R(\theta) \cdot P$$

$$= R(45^\circ) \cdot P$$

$$= \begin{bmatrix} \cos 45 & -\sin 45 & 0 \\ \sin 45 & \cos 45 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 8 & 14 \\ 3 & 10 \\ 1 & 1 \end{bmatrix}$$

$$\therefore P' = \begin{bmatrix} 3.5 & 2.8 \\ 7.8 & 14.0 \\ 1 & 1 \end{bmatrix}$$

~~Ans~~

b) About fixed point $(4,2)$.

i) Translate the fixed point to the origin i.e. $T(-4, -2)$

ii) Rotate by angle (θ) i.e. $R(45^\circ)$.

iii) Re-translate the fixed point to the original place.
i.e. $T(4, 2)$.

$$\therefore \text{COM} = T(4, 2) \cdot R(45^\circ) \cdot T(-4, -2)$$

Also,

$$P' = \text{COM} \cdot P$$

$$= T(4, 2) \cdot R(45) \cdot T(-4, -2) \cdot P$$

$$= \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos 45 & -\sin 45 & 0 \\ \sin 45 & \cos 45 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 8 & 14 \\ 3 & 10 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 2.1 & 1.4 \\ 3.5 & 2.7 \\ 1 & 1 \end{bmatrix}$$

$$\therefore P' = \begin{bmatrix} 6.1 & 5.4 \\ 5.5 & 14.8 \\ 1 & 1 \end{bmatrix}$$

Thus the rotated ~~at~~ points about 45° are:

$A(6.1, 5.5)$, $B(5.4, 14.8)$ $\underline{\underline{Ans}}$

2016-Spring

4. What will be the final co-ordinate of a polygon with vertices $A(3,4)$, $B(5,4)$, $C(5,2)$ & $D(3,2)$ after its reflected about a line $y = 2x + 1$.

Q. The given vertices of a polygon are:

$A(3,4)$, $B(5,4)$, $C(5,2)$ & $D(3,2)$.

The given line is: $y = mx + c$:

$$\therefore m = 2 \text{ \& } c = 1$$

$$\therefore \tan^{-1}(m) = \tan^{-1}(2) = 63.43$$

i) Translate the constant c to the origin i.e. $T(0, -1)$

ii) Rotate by angle $\tan^{-1}(m)$ i.e. $R^A(63.43)$

iii) Reflect about x -axis i.e. R_{fx}

iv) Re-rotate by angle $\tan^{-1}(m)$ i.e. $R^A(-63.43)$.

v) Re-translate the constant c to the original place.
i.e. $T(0, 1)$.

$$\therefore COM = T(0, 1) \cdot R^A(-63.43) \cdot R_{fx} \cdot R^A(63.43) \cdot T(0, -1)$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos(63.43) & -\sin(63.43) & 0 \\ \sin(63.43) & \cos(63.43) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \cos(63.43) & -\sin(63.43) & 0 \\ \sin(63.43) & \cos(63.43) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos(-63.43) & -\sin(-63.43) & 0 \\ \sin(-63.43) & \cos(-63.43) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.4 & -0.9 & -0.9 \\ -0.9 & -0.4 & 0.4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore \text{COM} = \begin{bmatrix} -0.6 & -0.8 & 0.8 \\ -0.8 & 0.6 & 0.4 \\ 0 & 0 & 1 \end{bmatrix}$$

Now

$$P' = \text{COM} \cdot P$$

$$= \begin{bmatrix} -0.6 & -0.8 & 0.8 \\ -0.8 & 0.6 & 0.4 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 3 & 5 & 5 & 3 \\ 4 & 4 & 2 & 2 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -0.6 \times 3 + -0.8 \times 4 + 0.8 \times 1 & -0.6 \times 5 + -0.8 \times 4 + 0.8 \times 1 & -0.6 \times 5 + -0.8 \times 2 + 0.8 \times 1 \\ -0.8 \times 3 + 0.6 \times 4 + 0.4 \times 1 & -0.8 \times 5 + 0.6 \times 4 + 0.4 \times 1 & -0.8 \times 5 + 0.6 \times 2 + 0.4 \times 1 \\ 0 \times 3 + 0 \times 4 + 1 \times 1 & 0 \times 5 + 0 \times 4 + 1 \times 1 & 0 \times 5 + 0 \times 2 + 1 \times 1 \end{bmatrix}$$

$$\begin{bmatrix} -0.6 \times 3 + -0.8 \times 4 + 0.8 \times 1 \\ -0.8 \times 3 + 0.6 \times 4 + 0.4 \times 1 \\ 0 \times 3 + 0 \times 2 + 1 \times 1 \end{bmatrix} = \begin{bmatrix} -4.2 & -5.4 & -3.8 \\ 0.4 & -1.2 & -2.4 \\ 1 & 1 & 1 \end{bmatrix}$$

Ans

2017-Spring

5. A mirror is placed vertically such it passes through the points $(10, 0)$ to $(0, 10)$. Find the reflected view of triangle ABC with co-ordinate $A(5, 50)$, $B(20, 40)$, and $C(10, 70)$.

sol Since a triangle ABC

Since a mirror is passes through the points $(10, 0)$ & $(0, 10)$. Then,

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$\Rightarrow y - 0 = \frac{10 - 0}{0 - 10} (x - 10)$$

$$\Rightarrow y = -1(x - 10)$$

$$\Rightarrow y = -x + 10$$

$$\therefore y = -x + 10$$

On comparing with $y = mx + c$, we get:-
 $m = -1$ & $c = 10$

To find the reflected view of triangle ABC with co-ordinates $A(5, 50)$, $B(20, 40)$ & $C(10, 70)$.

i> Translate the constant c to the origin i.e. $T(0, -10)$.

ii> Rotated by an angle $\tan^{-1}(m)$ i.e. $R(\tan^{-1}(-1)) = R(-45^\circ)$.

iii> Reflect about x -axis i.e. R_{fx} .

iv> Re-rotate by an angle $\tan^{-1}(m)$ i.e. $R^A(\tan^{-1}(-1)) = R^A(-45^\circ)$.

v> Re-translate the constant c to the original place i.e. $T(0, 10)$.

~~We know,~~

i.e. $COM = T(0, 10) \cdot R^A(45) R^A(-45) \cdot R(0, -10)$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos 45 & -\sin 45 & 0 \\ \sin 45 & \cos 45 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos(45) & -\sin(-45) & 0 \\ \sin(-45) & \cos(-45) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos 45 & -\sin 45 & 0 \\ \sin 45 & \cos 45 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0.7 & 0.7 & -7.1 \\ 0.7 & -0.7 & 7.1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore COM = \begin{bmatrix} 0 & 1 & -10 \\ 1 & 0 & 10 \\ 0 & 0 & 1 \end{bmatrix}$$

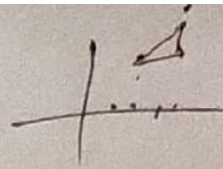
Now,

$$P' = COM \cdot P = \begin{bmatrix} 0 & 1 & -10 \\ 1 & 0 & 10 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 5 & 20 & 10 \\ 50 & 40 & 70 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\therefore P' = \begin{bmatrix} 40 & 30 & 60 \\ 15 & 30 & 20 \\ 1 & 1 & 1 \end{bmatrix}$$

~~Ans~~

2023 Spring



2.27 Reflect an object $(2,3), (4,3), (4,5)$ about line $y = x + 1$.

The given line is: $y = x + 1$.
on comparing with $y = mx + c$, we get;
 $m = 1$ & $c = 1$

$$\therefore \tan^{-1}(m) = \tan^{-1}(1) = 45^\circ$$

- i) Translate constant c to the origin i.e. $T(0, -1)$.
- ii) Rotate by an angle $\tan^{-1}(m)$ i.e. $R(45)$.
- iii) Reflect about x -axis i.e. R_x .
- iv) Re-rotate by an angle $[-\tan^{-1}(m)]$ i.e. $R(-45)$.
- v) Re-translate constant c to the original place i.e. $T(0, 1)$.

$$\text{i.e. COM} = T(0, 1) \cdot R^A(-45) \cdot R_x \cdot R^A(45) \cdot T(0, -1)$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(-45) & -\sin(-45) & 0 \\ \sin(-45) & \cos(-45) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \cos(45) & -\sin(45) & 0 \\ \sin(45) & \cos(45) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(-45) & -\sin(-45) & 0 \\ \sin(-45) & \cos(-45) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.7 & -0.7 & 0.7 \\ -0.7 & -0.7 & 0.7 \\ 0 & 0 & 1 \end{bmatrix}$$